

3. (a) Most of the tap water in Wales comes from mountain reservoirs. The water from these reservoirs is never pure.

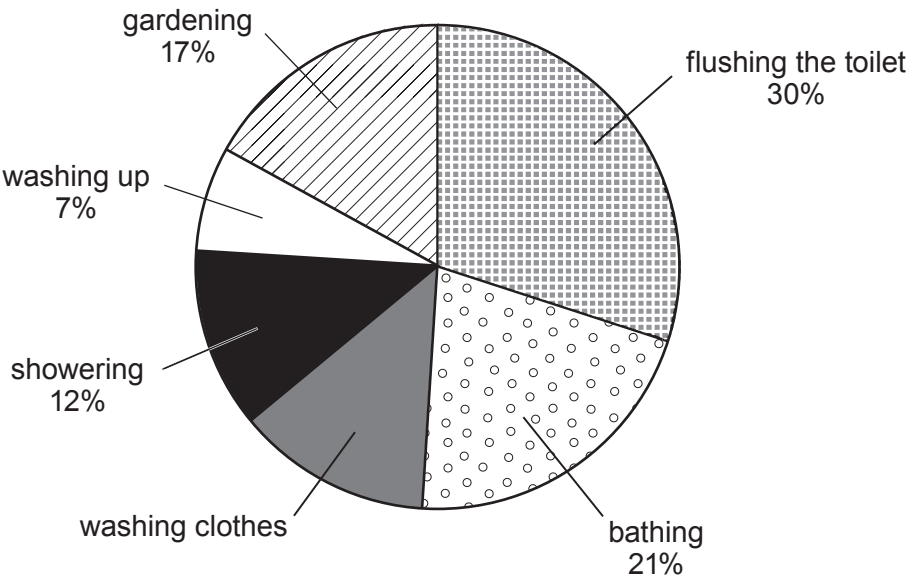
(i) Information about some of the stages used in the treatment of tap water is shown.

Draw a line to connect each stage to the correct description. One has already been completed. [2]

Stage	Description
filtration	large insoluble particles are allowed to settle out as sludge
screening	screens are used to remove objects such as leaves, sticks and litter
chlorination	harmful bacteria are killed to make the water safe to drink
sedimentation	small insoluble particles are removed by passing the water through sand beds



(ii) The pie chart shows some of the main uses of water in a typical Welsh home.



Use the pie chart to answer parts I and II.

I. Calculate the percentage of water used for washing clothes. [1]

..... %

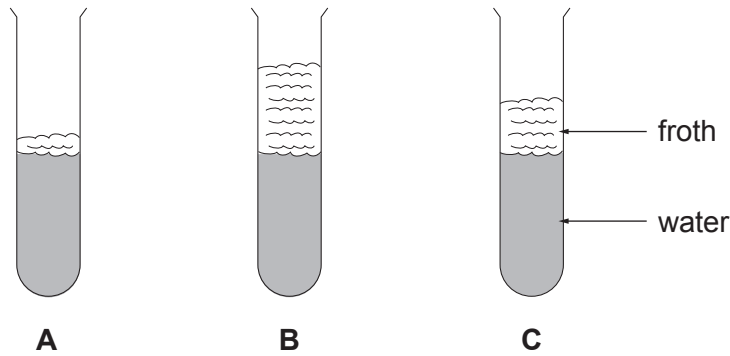
II. Give the name of the use that accounts for three times as much water as another use. [1]

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(b) 5 cm³ of three water samples, labelled **A**, **B** and **C**, were mixed with 10 drops of soap solution and shaken. One of the samples was distilled water. The boiling point of each water sample was also measured. The results obtained are shown.



Water sample	Boiling point (°C)
A	102
B	100
C	101

(i) Place the water samples, **A**, **B** and **C**, in order of hardness. [1]

hardest

.....

softest

(ii) Give the letter of the sample that is distilled water. Give **two** reasons for your choice. [2]

Sample

Reasons

1.
-
2.
-



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(iii) The volume of each water sample and soap solution was kept the same for each test. State which other variable must also be controlled before measuring the froth height. Describe how you would control this. [2]

.....

.....

.....

(iv) Hardness in water can be caused by dissolved calcium fluoride. Give the chemical formula for calcium fluoride. [1]

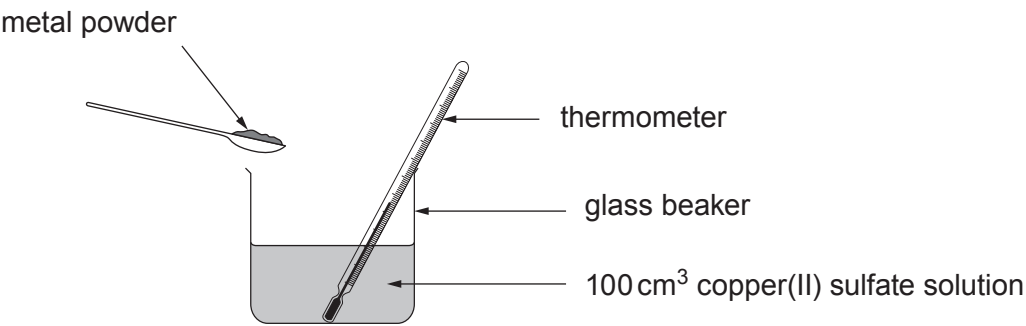
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3430U201
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3. Magnesium, zinc and iron powders were each added separately to 100 cm³ of copper(II) sulfate solution, to see which gave the greatest temperature change.



The temperature was recorded before and after each reaction. The results are shown in the table.

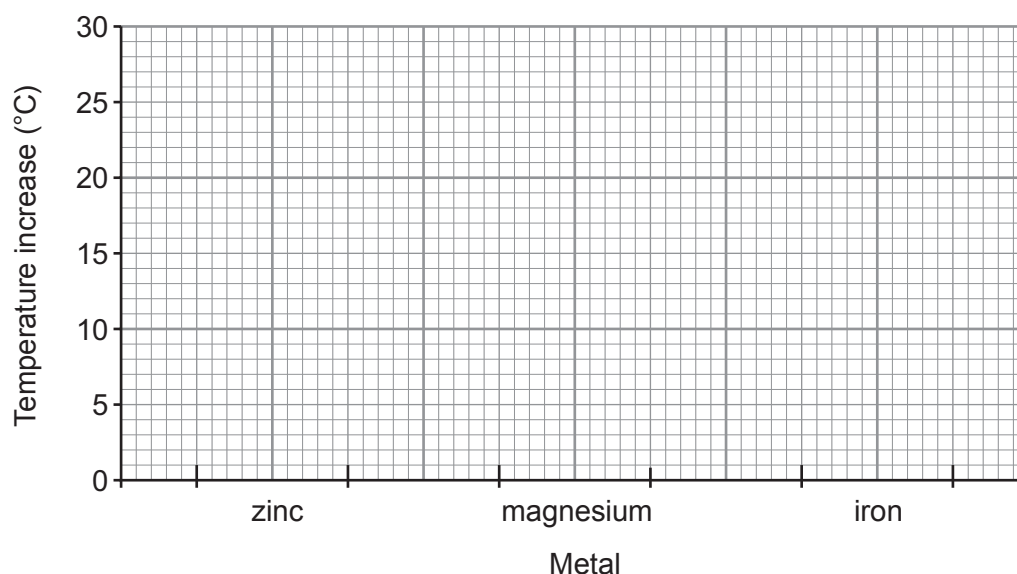
Metal	Temperature before the reaction (°C)	Temperature after the reaction (°C)	Temperature increase (°C)
zinc	20		14
magnesium	19	39	20
iron	19	24	5

- (a) Calculate the temperature after the reaction with zinc. [1]

Temperature = °C



- (b) Plot a **bar chart** to show the temperature increase for each metal. [2]



- (c) (i) Calculate the energy released in the reaction with magnesium. Use the following equation. [2]

$$\begin{array}{ccccc} \text{energy released} & = & \text{volume of solution} & \times & 4.2 \times \text{temperature increase} \\ \text{(J)} & & \text{(cm}^3\text{)} & & \text{(}^\circ\text{C)} \end{array}$$

Energy released = J

- (ii) In this reaction magnesium sulfate, MgSO_4 , is formed. What is the relative formula mass (M_r) of magnesium sulfate? [2]

$$A_r(\text{Mg}) = 24 \quad A_r(\text{S}) = 32 \quad A_r(\text{O}) = 16$$

M_r =

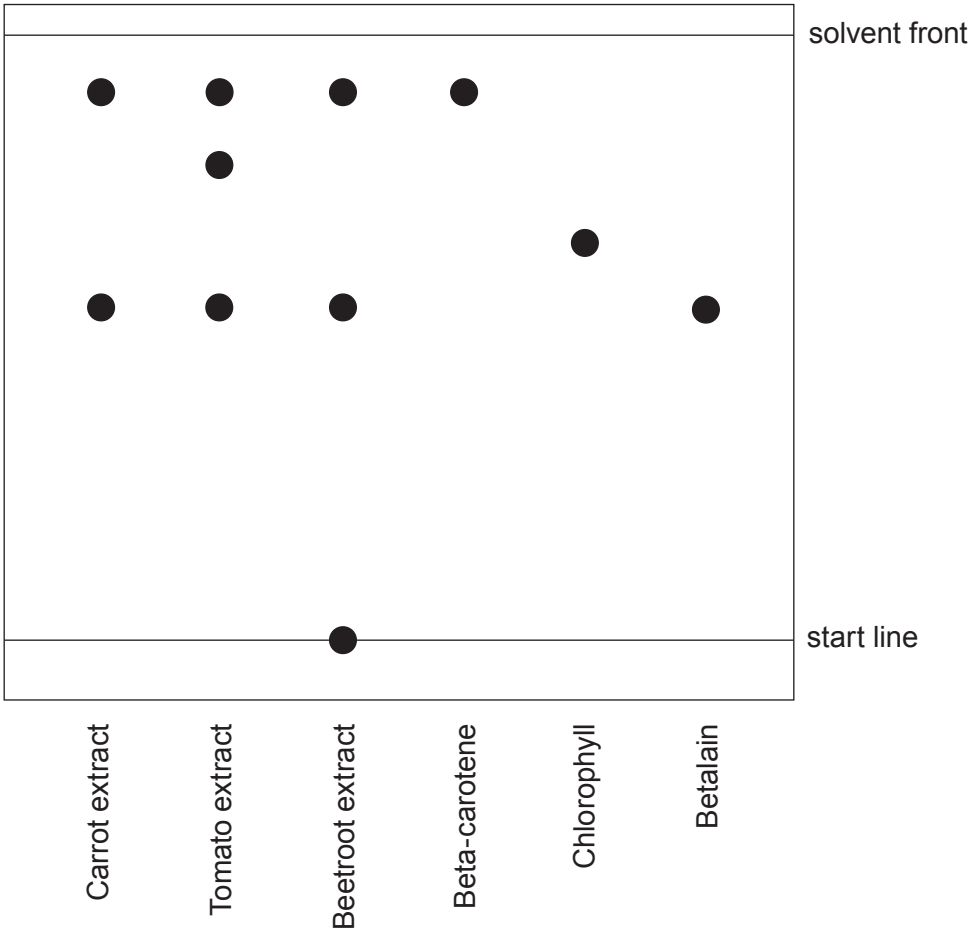


3. A student investigates the pigments found in some fruit and vegetables.

She obtains some coloured extracts from carrots, tomatoes and beetroot.

She places a spot of each extract on chromatography paper, along with spots of the three pigments beta-carotene, chlorophyll and betalain.

The diagram shows the chromatogram at the end of the experiment.



(a) Give the names of **all** the substances which are **not** mixtures. Give a reason for your choice. [1]

Substances

.....

Reason

.....



- (b) Put a tick (✓) in the boxes next to the **two** conclusions that can be drawn from the chromatogram. [2]

chlorophyll is not present in carrot, tomato or beetroot extracts

☐

beta-carotene is present in carrot extract but not present in tomato extract

☐

both beta-carotene and betalain are present in beetroot extract

☐

betalain is present in tomato extract but not present in carrot extract

☐

both carrot and beetroot extracts contain a pigment other than beta-carotene, chlorophyll and betalain

☐

- (c) One of the pigments present in the carrot extract has travelled 4.4 cm above the start line.

The solvent front has travelled 8.0 cm. Calculate the R_f value of the pigment using the following equation. [2]

$$R_f = \frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent front}}$$

$R_f =$

- (d) Give the reason why there is a spot remaining on the start line in the chromatogram for beetroot. [1]

.....

.....



3. A beryllium atom can be represented by the following symbol.



- (a) Choose a number from the box to complete each of the following sentences. [4]

4	5	9	13
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Each number can be used once, more than once or not at all.

Beryllium has protons.

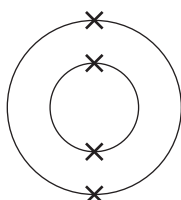
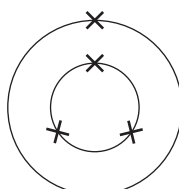
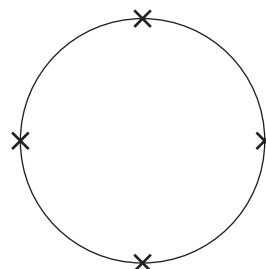
The atomic number of beryllium is

The mass number of beryllium is

Beryllium has neutrons.

- (b) Beryllium has 4 electrons.

Put a tick (✓) in the box that shows the correct electronic structure for beryllium. [1]


☐

☐

☐

- (c) Beryllium is found in the compound beryllium fluoride, BeF_2 .

Give the total number of atoms shown in the formula. [1]

.....



3. The table shows how the solubility of potassium nitrate in water varies with temperature.

Temperature (°C)	Solubility (g per 100g of water)
0	13
10	21
20	32
30	46
40	64
50	86

(a) Tick (✓) the box that shows approximately how many times more soluble potassium nitrate is at 40 °C than it is at 10 °C.

[1]

2 times

☐

3 times

☐

4 times

☐

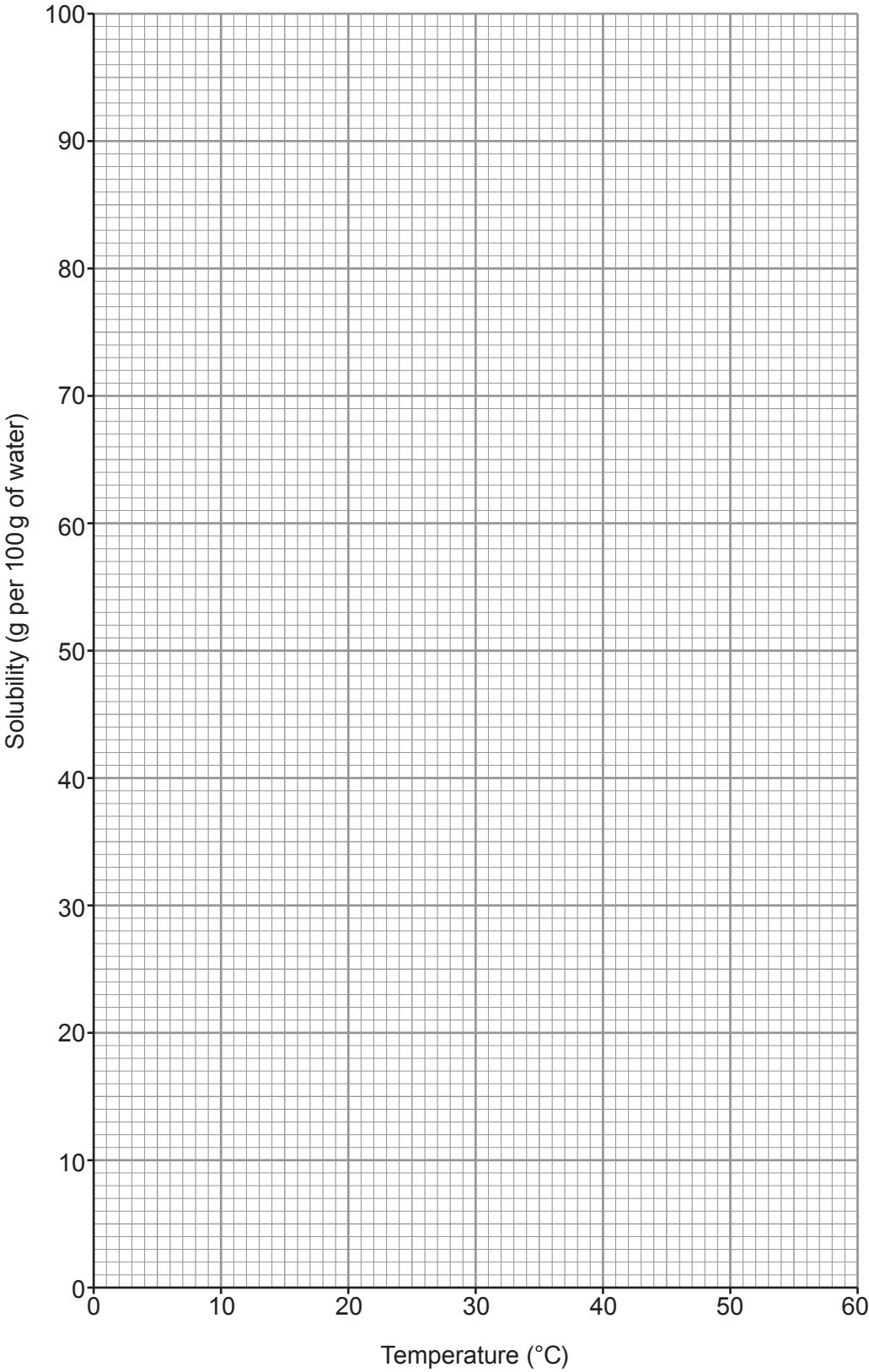
5 times

☐

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- (b) (i) Plot the solubility of potassium nitrate against temperature on the grid.
Draw a suitable line.

[3]



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- (ii) I. State the solubility of potassium nitrate at 45 °C. [1]

..... g per 100 g of water

- II. Calculate the mass of solid potassium nitrate that forms when a saturated solution in 100 g of water at 45 °C cools to 25 °C. [1]

Mass = g

- (c) Potassium nitrate decomposes on heating to produce potassium nitrite, KNO_2 , and oxygen.

- (i) Balance the equation for the reaction. [1]



- (ii) On heating potassium nitrate, a student expected to make 4.35 g of potassium nitrite. However, in an experiment she only made 3.13 g.

Calculate the percentage yield of potassium nitrite in her experiment. Give your answer to the nearest **whole** number. [2]

Percentage yield = %

